

$\langle u, v \rangle = u^T A v$ $\langle ; \rangle$ je skalarný součin
 iff $a_{11} > 0$ a $\det(A) > 0$ a A symetrická

$$u = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}; \quad v = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

SSS

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} a_{11} x_1 + a_{21} x_2 & a_{12} x_1 + a_{22} x_2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

②

$$= a_{11} x_1 y_1 + a_{21} x_2 y_1 + a_{12} y_1 x_1 + a_{22} x_2 y_2$$

$$\begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} y_1 + a_{21} y_2 & a_{12} y_1 + a_{22} y_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

①

$$= a_{11} x_1 y_1 + a_{21} x_2 y_1 + a_{12} y_1 x_2 + a_{22} y_2 x_2 \quad \text{ale protože je symetrická}$$

tak $a_{12} = a_{21}$ a tedy i ① = ②

lineární kombinace
máto

SL1 $\vec{u}^T A (k \vec{v}) = \langle \vec{u}; k \vec{v} \rangle = \vec{u}^T A (k \vec{v}) =$

$$= k \vec{u}^T A \vec{v} = k \langle \vec{u}; \vec{v} \rangle \quad \text{OK}$$

SL2 $\langle \vec{u}; \vec{v} + \vec{w} \rangle = \vec{u}^T A (\vec{v} + \vec{w}) = \vec{u}^T A \vec{v} + \vec{u}^T A \vec{w} =$

$$= \langle \vec{u}; \vec{v} \rangle + \langle \vec{u}; \vec{w} \rangle \quad \text{OK}$$

SPR

